

modeller is that there is a tweak tool, which can be used to reshape and adjust the look and proportions of the 3D model even after it has been completed.

The virtual mirror tool—a great boon for organic modelling

A virtual mirroring tool is one that organic artists hunger for; note that even popular 3D software like Realsoft3D do not have symmetrical modelling or a virtual mirroring tool. The virtual mirror tool helps artists to create their models symmetrically. You only need to work on one side, and the other side is taken care of by the virtual modeller. Mirroring is possible on any axis—X, Y or Z. The virtual mirror tool mirrors the object opposite the face selected

Key	Function
c	Connect selected vertexes or edges
d	Repeat last operation
Ctrl+z	Toggle undo/redo
Shift+Ctrl+z	Redo previous operations (multiple redo possible)
Alt+Ctrl+z	Undo previous operations (multiple undo possible)
Numeric keys on top of keyboard 2 -> 0	Cut selected edge into 2 to 10 edges.

Table 3: Editing keys

for operation. The greatest advantage of the virtual mirror is that it is a great time-saver and helps in creating organic models of animals and human beings.

To mirror an object, go to the Face mode by pressing the *F* key on the keyboard. Click on Face with the LMB. Go to the Tools menu, click the *Virtual*

Mirror sub-menu, and then the first entry, titled *Create*. After completing the model, click the *Freeze* option (the third under the *Virtual Mirror* sub-menu). This operation creates a symmetrical model ready for exporting to other formats like object and 3ds. If you are not satisfied with the mirroring, you can even break the symmetry by selecting the second option under the *Virtual Mirror* sub-menu.

In the present case, to avoid delay, we would like to illustrate free-form modelling instead of the traditional exact way of bringing in an image into the image plane, to be displayed as a guide in the background. We will also avoid the virtual mirror tool, since it is not required at the present juncture—I am going to illustrate a simpler technique of creating a model of the airplane in no time. For those of you interested in modelling an airplane exactly, go to the website the-blueprints.com and download the blueprint of your choice. Size it in the GIMP, and import it into the image plane.

Modelling the airplane

Look at Figures 3 to 5. First click the RMB. From the menu, select a cube, and with a left click, select the face. Press the *F* key if you are not in *Face selection* mode.

Click the RMB (see Figure 6) and choose *Extrude Normal* (see Figure 7) six times, one after the other, to get to Figure 8.

Now go to *Edge selection* mode (press the *E* key, or click the icon as seen in Figure 9). Select a top edge (see Figure 10), press the *L* key to select the whole edge loop, and scale it down uniformly (see Figure 11).

Selecting the edge loops, go on scaling up or down the figure, till it gets

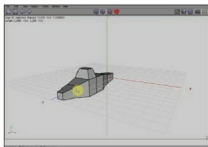


Figure 18: Opposite side edges scaled out

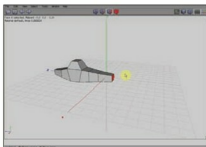


Figure 21: Back face selected

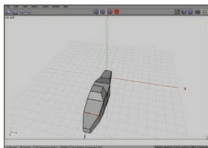


Figure 19: Second edge loop scaled in X axis to shape the front part

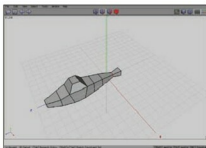


Figure 22: Penultimate edge loop scaled down

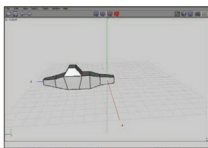


Figure 20: Shifting edges to shape the cockpit

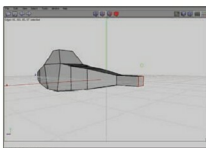


Figure 23: Last edge loop scaled down